

Technology Stack

Date	02 July 2026
Team ID	SWTID-2026-8076
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Python and Scikit-learn are used to build and train the Machine Learning model for credit card approval prediction. Flask connects the trained model with the web application, while HTML and CSS provide a simple and user-friendly interface. Pandas and NumPy are used for data preprocessing, and Matplotlib is used for data visualization. Git and GitHub are used for version control and project management. Together, these technologies provide a reliable, maintainable, and efficient Credit Card Approval Prediction System.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	Used to create a simple, responsive, and user-friendly interface for entering applicant details and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Python provides Machine Learning support, while Flask integrates the trained model with the web application and handles prediction requests.
3	Database / Data Storage	CSV Dataset	Stores the applicant data used for model training and testing.
4	Development Environment	Jupyter Notebook, VS Code	Jupyter Notebook is used for data analysis and model training, while VS Code is used for Flask application development.
5	Version Control & CI/CD	Git, GitHub	Used for source code management, version tracking, collaboration, and project backup.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib	Scikit-learn is used for model training and prediction. Pandas and NumPy are used for pre-processing and analysis, while Matplotlib is used for visualization.